

An FPGA-Based Time-to-Digital Converter With Online Dual-Chain Calibration

Zhengsen Jia^{1b}, Yuzhuo Wang^{1b}, Jie Ding^{1b}, Qian Xu^{1b}, and Yadongyang Zhu^{1b}

Abstract—This article introduces a field-programmable gate array (FPGA)-based time-to-digital converter (TDC) with an online dual-chain calibration method to address time resolution and linearity issues arising from the non-uniformity of FPGA internal delay elements. The aim is to enhance the stability and accuracy of TDCs under varying environmental conditions. We employ an external high-precision clock to generate start-stop calibration signals for real-time monitoring of signal propagation time changes within FPGA components. Two identical delay chains in the TDC are managed by a finite state machine (FSM) to switch their roles, ensuring continuous data acquisition and online calibration. Experimental results show that the proposed TDC significantly outperforms traditional time interval counter (TIC) in terms of time linearity, maximum integral nonlinearity (INL), and differential nonlinearity (DNL). Within a temperature range of 20 °C–80 °C, post-online temperature calibration maintains the root mean square (rms) of the TDC below 20 ps, substantially improving performance across different temperature conditions.

Index Terms—Field-programmable gate array (FPGA), online calibration, temperature compensation, time delay line (TDL), time interval measurement, time-to-digital converter (TDC).

I. INTRODUCTION

IN THE fields of scientific research and engineering technology, time-to-digital converters (TDCs) are essential components for time interval counters (TICs), playing an indispensable role in precise time measurement and analysis. They are essential in various applications, such as, time and frequency metrology, Global Navigation Satellite System (GNSS) [1], [2], [3], [4], light detection and ranging [5], [6], [7], time-of-flight positron emission tomography [8], [9], [10], [11], [12], [13], and high-energy physics experiments. In recent decades, field-programmable gate array (FPGA)-based TDCs have garnered significant attention due to their design flexibility, and cost-effectiveness [14], [15], [16], [17], [18], [19], [20], [21]. These designs typically employ coarse counters operating at nanosecond-level system clocks to expand the

range, coupled with finer time interpolation methods for higher resolution. The use of fast carry chains (CARRY) in FPGAs has made delay-line-based TDCs the dominant implementation. Common TDC methods include basic time-delay-line (TDL) techniques, ring delay chain methods, Vernier delay chain methods, and wave union (WU) methods. However, FPGA-based TDL-TDCs still face challenges in time resolution and linearity, particularly due to the non-uniformity of delay elements caused by specific components used in FPGAs, which can disrupt time linearity.

Traditional TDCs, using only the rising or falling edge of a trigger signal in single measurements, are limited by the inherent delay of CARRYs. To overcome this limitation, Wu and Shi [22] proposed a CARRY-based finite waveform generator in 2008, which converts single-edge trigger signals into dual-edge pulse trains, significantly enhancing TDC resolution through multiple measurements. Later advancements produced generators capable of generating pulse trains with four, six, and eight edges [23], [24], [25]. Additionally, a resource-efficient pattern generator was introduced, generating limited-edge signals with minimal look-up tables (LUTs) and buffers, offering a resource-friendly solution despite its shorter operational range [23], [26]. To address the short operational range, Wu and Shi [22] further proposed an infinite waveform generator based on ring oscillators, capable of producing clock-like signals spanning multiple clock cycles, though potentially affecting stability due to sparse edge arrangements [27]. Recently, the multi-sampling waveform union method was introduced, utilizing LUT-based waveform generators to achieve a wider operational range and generate low-jitter pulse trains, albeit at the cost of increased resource consumption to mitigate pulse stretching effects [28].

In TDL-TDCs, the number of chains is a critical factor. Multi-TDL architectures enhance interpolation density, supporting multiple measurements with extensive logical resources [14], [29]. The length of TDLs is also crucial, as an increase in the number of delay elements introduces more measurement uncertainty. To address this, phase-locked loops (PLLs) are commonly used in FPGAs to boost the system clock frequency, maintaining shorter TDL lengths. Additionally, in waveform union methods, the width of pulse trains is considered part of their total length.

With advancements in FPGA manufacturing technology, the issue of bubbles in FPGA-TDCs has become more pronounced. For single-measurement or single-edge per sample scenarios, a single counter encoder offers a simple and efficient solution, employing global bubble error correction to

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automatically sort bin order [14], [27], [29]. However, bubble removal becomes more complex in waveform union methods. In 45 nm FPGA-TDCs (e.g., Spartan-6), bubbles are relatively small and can be handled through enumeration and direct removal [23], [30]. In advanced 28 nm FPGAs, the issue is exacerbated as TDL locations span across two adjacent clock regions. Although bin alignment methods can mitigate bubble effects [30], the bin sorting dynamically changes due to the process, voltage supply, and temperature (PVT), making complete bubble elimination challenging. A time-saving approach involves decomposing the original code into several subsegments and using perfect thermometer codes. To prevent pulse train distortion, extending TDLs to accommodate wider pulse trains is necessary, which consumes more logical resources [29].

To effectively monitor and respond to temperature fluctuations in FPGAs, this study introduces an online calibration method. This method leverages an external high-precision clock to generate start-stop calibration signals, enabling real-time monitoring of signal propagation time changes within the FPGA's internal components. The TDC employs two identical delay chains that alternate roles as the primary and standby delay chains. The precise management of the finite state machine (FSM) ensures effective control over the switching between the calibration delay chain and the standby delay chain.

During initialization, the FSM selects one delay chain for code density testing. Upon completion, this chain is designated as the primary working chain, responsible for responding to external trigger signals, while the other serves as the standby. During normal operation, calibration signals start and stop the counting process of a ring oscillator to measure oscillation frequency over a specified duration. If frequency variations exceed a preset threshold, the system automatically triggers the standby chain for code density testing to recalibrate it. After calibration, the system switches the roles of the two chains to ensure continuous data acquisition and online compensation.

Experimental results demonstrate that the proposed method maintains high performance even in variable temperature environments compared to traditional TICs. This innovation ensures that the proposed TDC exhibits excellent stability and accuracy in time difference measurement under diverse external conditions.

This article is an extension of the proceedings abstract published at the 2024 Conference on Precision Electromagnetic Measurements (CPEM 2024) [31]. Compared to the abstract, this article introduces improvements by incorporating a signal conditioning circuit to ensure that the signal amplitude and rising edge fed into the FPGA meet the required standards. The delay chain of the FPGA has been redesigned, and constraints have been optimized to ensure the uniformity and consistency of CARRY4 and flip-flops. Due to changes in the TDC circuit and internal constraints, the standard deviation and root mean square (rms) under different temperatures were retested. Furthermore, a comparative experiment between the TDC and SR620 was conducted, including comparisons of background noise under temperature variations and fixed delay measurement results.

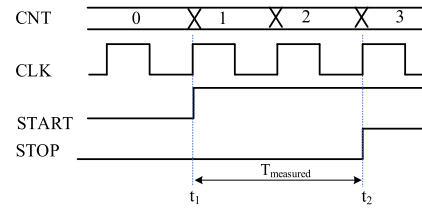


Fig. 1. Time-interval measurement with a coarse-counter TDC.

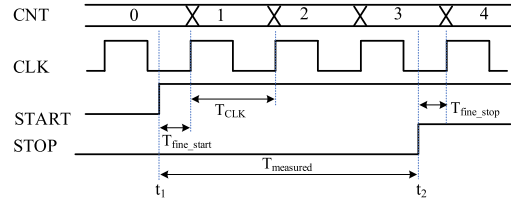


Fig. 2. Time-interval measurement with a coarse-fine counter TDC.

II. COARSE-FINE COUNTING PRINCIPLE

The direct counting principle is a fundamental method in precise time interval measurement, suitable for a wide range of measurements but with lower resolution. As shown in Fig. 1, this method involves measuring the time difference between external START and STOP signals. In an ideal scenario, if the clock signal (CLK) used for sampling the input signals has no delay, the system clock's period can directly count the number of complete clock cycles within the time interval, enabling rapid time interval measurement. However, in practical applications, the rising edges of START and STOP signals often do not perfectly synchronize with the clock signal, potentially leading to significant errors in the measurement results. Additionally, due to imprecise signal alignment, the measurement system may have a dead zone of up to two clock cycles, further affecting the accuracy and reliability of the measurements.

A method employs a strategy that combines coarse and fine counting to correct measurement errors is used, as depicted in Fig. 2. This method involves two measurement steps.

- 1) *Coarse Counting*: This step calculates the number of clock pulse rising edges that occur between the START and STOP signals, so that the coarse count time interval T_{coarse} can be calculated. For this relatively large time span, we simply track the complete clock cycles.
- 2) *Fine Counting*: This involves measuring the time intervals from the occurrence of the START signal to the subsequent clock pulse rising edge ($T_{\text{fine_start}}$) and from the STOP signal to the following rising edge of clock ($T_{\text{fine_stop}}$). This requires a more precise timing mechanism to capture these brief time intervals.

Thus, the measured time interval (T_{measured}) is calculated as a combination of the following three components:

$$T_{\text{measured}} = T_{\text{coarse}} + T_{\text{fine_start}} - T_{\text{fine_stop}} \quad (1)$$

where $T_{\text{coarse}} = T_{\text{CLK}} \times N$ is the coarse counting time interval. N represents the number of passed complete CLK cycles, and T_{CLK} is the fixed period of the clock signal.

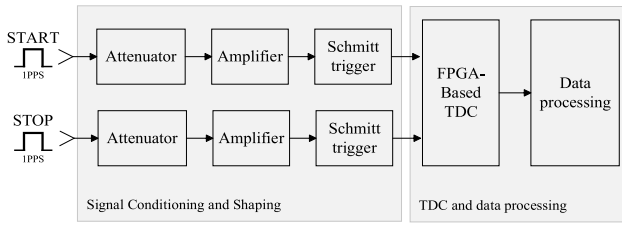


Fig. 3. Architecture of proposed TIC.

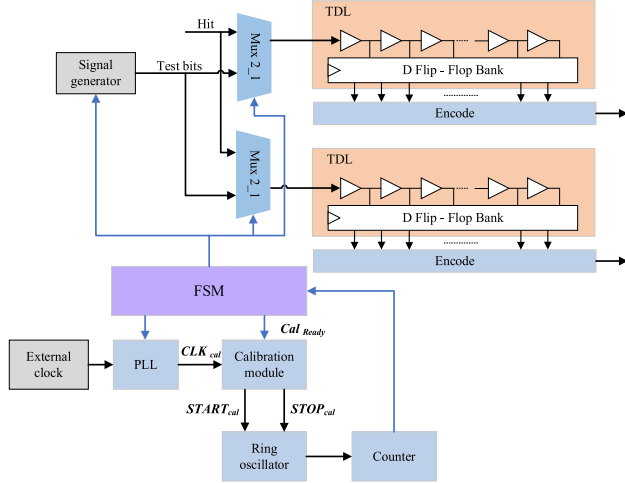


Fig. 4. Architecture of the proposed TDC measurement system.

III. PROPOSED TIC

A. Architecture of Proposed TIC

Proposed TIC mainly consists of two parts, one is signal conditioning and shaping, the other one is time interval counting. Its architecture is shown in Fig. 3.

To ensure that the 1PPS signal input to the FPGA meets the counter's requirements and minimizes errors, a signal conditioning circuit is implemented at the front end. This signal conditioning and shaping circuit is designed to ensure that the 1PPS signal satisfies the FPGA's high and low logic level requirements for input signals, that is, when the input voltage is greater than or equal to 2 V, the FPGA considers the signal as high level, and when it is less than or equal to 0.7 V, it is considered low level, while also ensuring that the rise time is less than 2 ns. To achieve this, the ADCMP572 is used as the front-end signal conditioning circuit, which has a typical propagation delay of 150 ps and a high input bandwidth of 8 GHz, ensuring that the signal transitions quickly without distortion. The core component of this design is the TDC based on the FPGA, which features both coarse and fine counting. Additionally, digital filtering is incorporated into the data processing to eliminate gross measurement errors.

B. Proposed FPGA-Based TDC Module

Fig. 4 illustrates the architecture of the TDC module proposed in this article, which integrates two TDLs composed of CARRY4 units, one for time interval measurement and the other for standby. When the circuit delays changes, the standby TDL is switched to the work chain after code density test.

The operation states of the TDL are controlled by an FSM. This FSM controls the compensation module to measure the oscillation frequency of the ring oscillator, determining

whether the performance of the delay elements in the current circuit has changed. Based on the determination results, the state machine utilizes control signals to switch the operation and calibration states of the two TDLs, as well as to manage transitions between states.

CARRY4 units in an FPGA are specialized components designed for high-speed arithmetic operations. Each CARRY4 contains four carry chains, which can also serve as delay units. This configuration provides consistent propagation delays, making CARRY4 ideal for precise delay chains necessary in high-resolution time measurement applications.

To enhance system accuracy and reliability, the design of the delay chain is crucial. By carefully constraining the placement of CARRY4 units in the FPGA using Xilinx design constraints files within Vivado and utilizing tool command language scripts, uniformity and consistency in the delay chain are achieved. Initially, the location constraint places the first CARRY4 unit in a specific configurable logic block (CLB), clearly defining the starting point of the delay chain. Following this, relative location constraints arrange the subsequent CARRY4 units consecutively within the same CLB column. This meticulous layout ensures consistent delays across each unit, facilitating high time resolution.

Furthermore, flip-flops that latch the carry chain data are strategically positioned near the CARRY4 units. This proximity minimizes data path length, ensuring reliable sampling of carry chain data within the clock domain. These constraints collectively optimize delay chain performance, reduce fluctuations in network delay, and enhance overall system reliability and efficiency.

C. Code Density Test

Due to variations in PVT, the actual delays of the delay elements may differ. This inconsistency can result in integral nonlinearity (INL) and differential nonlinearity (DNL), which in turn affect the linearity of the TDC. The DNL and INL for the i th tap can be defined as follows:

$$DNL_i = \frac{W_i - W_{LSB}}{W_{LSB}} \quad (2)$$

$$INL_i = \sum_{k=1}^i DNL_k. \quad (3)$$

To estimate the actual bin width of each delay element, the code density test is employed. This method calculates each delay element's bin width W_i by counting the number of hits N_i at each trigger point, thus assessing the performance of the delay elements. In this design, a sample size $N = 1\,024\,000$ is selected to balance resource usage and correction effectiveness. The bin width W_i for each delay element can be determined by the following equation:

$$W_i = \frac{N_i}{N} T_{clk}. \quad (4)$$

Once the actual bin widths of each delay element are obtained using code density testing, INL can be calibrated. Wu proposed a bin-by-bin calibration structure, where DNL and INL values are derived from histograms and used for bin-by-bin calibration [5]. Wu suggests correcting the center of

TABLE I

FPGA RESOURCE UTILIZATION AND POWER CONSUMPTION					
Resources	Available	Single-Chain		Dual-Chain	
		Used	Utilization	Used	Utilization
LUT	133,800	1223	0.91%	2575	1.92%
CLB Flip-Flop	269,200	821	0.30%	1847	0.69%
Block RAM	365	2	0.05%	4	1.10%
Total Power		1.521W		1.619W	

chain for a code density test to ensure accurate measurement of signal propagation time. Upon successful completion of the test, this chain is designated as the main working chain, and the FSM transitions to the work state. In this state, the working chain responds to external trigger signals and performs real-time measurements, while the other chain remains on standby. As long as the main working chain operates normally, the FSM remains in the work state, periodically moving to the monitor state.

In the monitor state, the calibration signal Cal_{Ready} controls the counting of the ring oscillator to measure its frequency over a set time, comparing the current frequency with the initial measurement. If the frequency change falls within the preset threshold, the FSM returns to the work state. If the frequency deviation exceeds the threshold, the FSM transitions to the calibration state.

In the calibration state, the system automatically activates the standby chain for a code density test and calibration to ensure accurate measurements under the new conditions. After calibration, the system swaps the roles of the two chains: the standby chain becomes the new working chain, while the original working chain switches to standby. The FSM then returns to the work state, ensuring continuous data collection and system calibration.

This FSM management approach, with automatic calibration and delay chain switching, allows the system to precisely monitor and respond to fluctuations in FPGA operating temperature, maintaining high performance and stability under varying thermal conditions.

IV. FPGA-BASED TDC PERFORMANCE TESTING

A. Experiment Setup

The TDC proposed in this article has been successfully implemented on a Xilinx Artix-7 200T FPGA. The resource utilization is presented in Table I. To generate external clock signals for measuring time intervals, a Si5338 EVB board manufactured by Silicon Labs is employed. During the design and implementation of the TDC, the Xilinx Vivado 2019.2 tool is used for synthesis, place and route, and generation of the corresponding bitstream. The design language for the TDC circuitry is Verilog hardware description language (HDL). The FPGA is integrated into a development board equipped with an RS232 serial communication interface, which is used to transmit all test results via serial port to a personal computer (PC). The input 1PPS reference signal is the same source as the FPGA time base and is connected to the input channel through the subminiature version A (SMA) connector to ensure stable signal input.

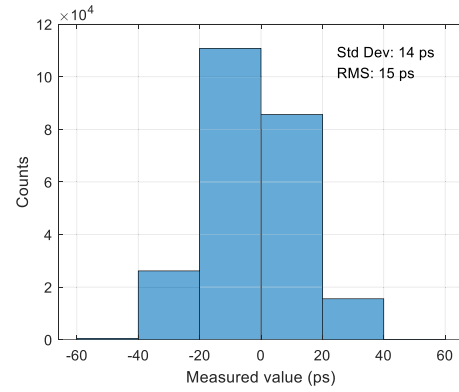


Fig. 8. Distribution of time interval measurements.

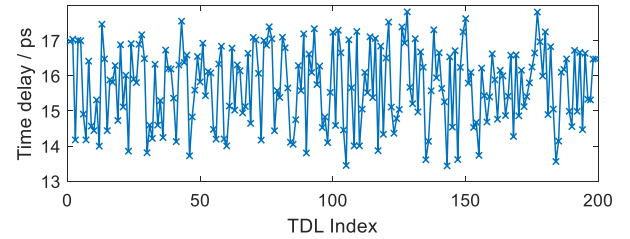


Fig. 9. Time delay of each delay element.

This article presents a detailed evaluation of the TDC's accuracy by fine-tuning the time intervals of different input 1PPS signals and varying the ambient temperature. This approach not only enhances the flexibility of the testing process but also significantly improves the accuracy and reliability of the measurement results.

B. Measurement Accuracy Analysis

To evaluate the accuracy performance of the TDC following code density testing, this study employed the FPGA to measure the zero-time interval value over 200 000 times without any data processing. The standard deviation and rms of the measurement results were calculated. Additionally, a bar chart depicting the distribution of different measurement value intervals was created, as shown in Fig. 8.

From Fig. 8, we can observe that 82.31% of the measured probability values are less than 20 ps. The standard deviation is 14 ps, and the rms is 15 ps. These results indicate that the effects of DNL and INL have been significantly improved, resulting in highly stable measurement outcomes.

The time delay of each delay element is calculated based on the code density test data, as shown in Fig. 9. Then, the LSB, DNL, and INL are calculated, the LSB is 15.7 ps, the DNL is less than 14.4%, and the INL is less than 30.4%. We compared our accuracy values with those reported in references [32], [33], [34], and the comparison results are shown in Table II.

The experimental results show that the TDC designed in this article achieves a balance between time resolution and nonlinear errors at the FPGA 28 nm process node. The LSB of the proposed method is 15.7 ps, which is a 31.7% improvement compared to [33] (23 ps) and an 89.9% improvement compared to [34] (156 ps). Although [32] has an advantage in terms of LSB (9.6 ps), its DNL and INL are as high as 68.3% and 683.0%, respectively, significantly higher than the

TABLE II
COMPARISON WITH OTHER METHODS

Reference	LSB (ps)	RMS/STD (ps)	DNL/INL (%)	TECH. Node (nm)
[32]	0.4	12 / -	68.3 / 683.0	28
[33]	23	-	100.0 / -	90
[34]	156	68 / -	23.3 / 26.4	28
Our method	15.7	15 / 14	14.4 / 30.4	28

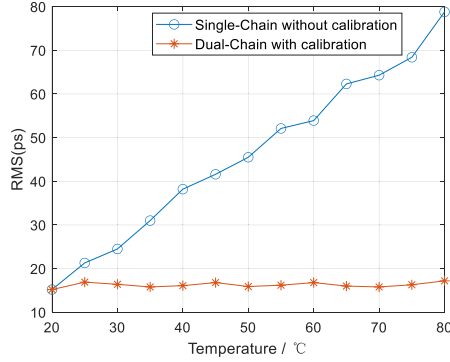


Fig. 10. rms of TDC under different temperature conditions.

14.4% and 30.4% of this article. A large INL ($>683\%$) can lead to significant error accumulation in long-time interval measurements, severely affecting the accuracy and long-term stability of the system. This comparison demonstrates that the method proposed in this article effectively controls nonlinear errors while ensuring high time resolution, achieving a balance between measurement accuracy and system stability.

C. Effects of Temperature Variation on TDC

To further investigate the impact of temperature changes on the delay of electronic components within the FPGA, this article calculates the rms of the zero-time interval values measured by the TDC across an ambient temperature range of 20 °C–80 °C. The results are presented in Fig. 10. The blue circular curve represents the rms results of a single delay chain without calibration, while the red star curve indicates the rms results of a dual delay chain with calibration.

The rms of the time interval values measured by the TDC using a single delay chain without calibration increase linearly with temperature. However, when employing two delay chains with calibration for time interval measurements, the rms remains consistently below 20 ps, even in environments with significant temperature fluctuations. This demonstrates a substantial improvement in performance across varying temperature conditions.

V. COMPARISON EXPERIMENT

A. Local Noise Testing

To verify the performance of the proposed TDC, this article presents a local noise comparison experiment using the SR620, a high-accuracy device widely used in time and frequency metrology. The experiment records the impact of temperature fluctuations on the local noise of the TDC. The measurement

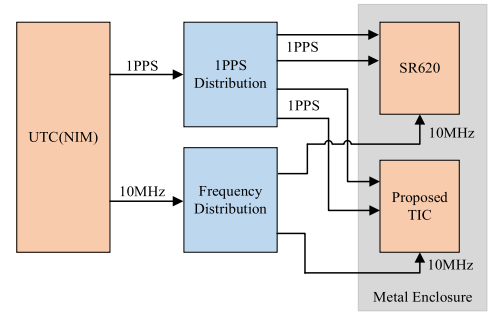


Fig. 11. Zero-time delay measurement scheme.

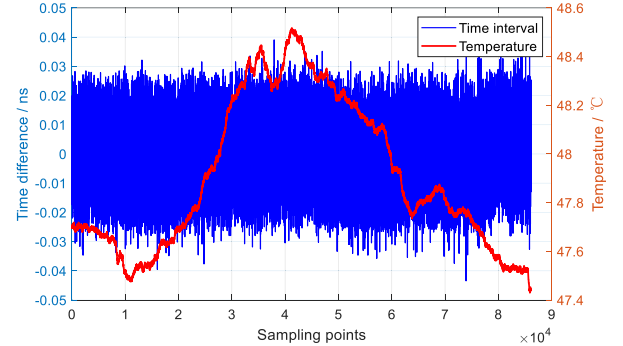


Fig. 12. Time difference of the proposed TDC versus temperature fluctuation.

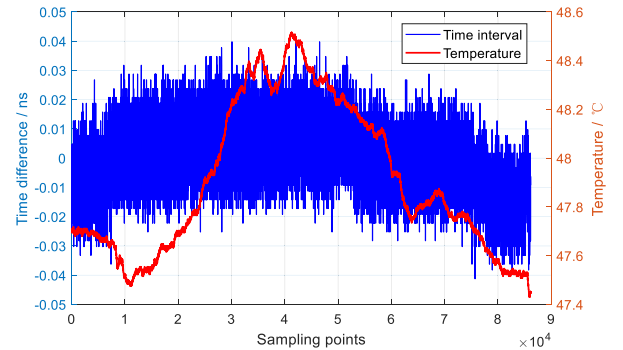


Fig. 13. Time difference of SR620 versus temperature fluctuation.

setup is shown in Fig. 11, where both devices are placed inside the same metal enclosure.

The signal source utilized in this article is a 1PPS output from UTC(NIM). The 1PPS signal is distributed into multiple outputs using a pulse distributor, with two outputs connected to the START and STOP input channels of a counter via coaxial cables of equal length. Simultaneously, two corresponding coaxial cables link the 1PPS output to the input of the SR620. Each device records a time difference value every second over a measurement period of one day while also monitoring its internal ambient temperature variations throughout the day. The resulting curves depicting the local noise of these two devices as a function of temperature are shown in Figs. 12 and 13.

As shown in Figs. 12 and 13, the time difference measured by the proposed TDC and the SR620 fluctuates within ± 30 ps, with the standard deviation of the proposed method being 9 ps, compared to 11 ps for the SR620. This indicates that both time interval measurement devices exhibit strong noise suppression capabilities. Notably, when the measurement environment's

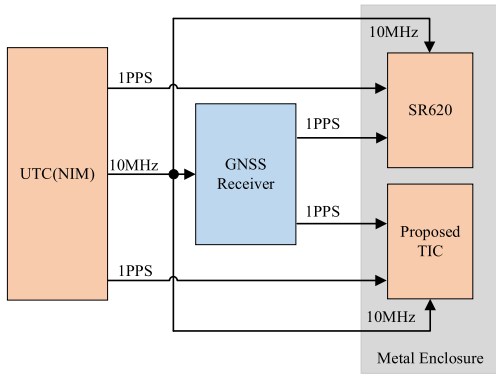


Fig. 14. Fixed time delay measurement scheme.

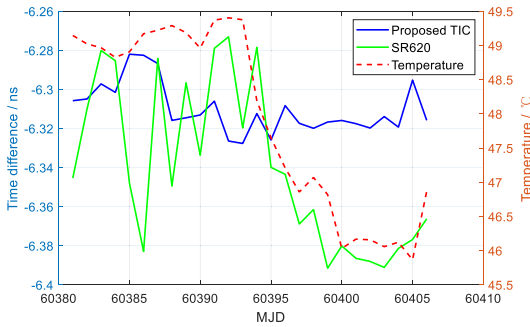


Fig. 15. Time difference of the proposed method versus temperature fluctuation.

temperature varies by $\pm 1^\circ\text{C}$, the time difference recorded by the proposed method remains relatively stable. These results demonstrate that the proposed method, which utilizes online calibration of code density, effectively mitigates the impact of temperature fluctuations on the measured time interval.

B. Fixed Time Interval Measurement

The 10 MHz output from the UTC(NIM) is connected to the GNSS receiver, where the internal crystal oscillator is synchronized to the external 10 MHz signal using a PLLs. The output 1PPS from the GNSS receiver originates from the same source as the 1PPS signal of the UTC(NIM), but the 1PPS signal passing through the GNSS receiver incurs a fixed delay. The proposed method was used to measure this fixed delay from MJD 60381 to MJD 60406, and the fixed time delay measurement scheme is illustrated in Fig. 14. The resulting curve showing the measured time difference as a function of temperature change is presented in Fig. 15.

The average time difference measured by the SR620 between MJD 60381 and MJD 60406 is -6.34 ns , with a standard deviation of 42 ps. In comparison, the average time difference measured by the proposed method is -6.31 ns , with a standard deviation of 13 ps. When the temperature varies from 45.5°C to 49.5°C , the time difference fluctuation remains within $\pm 50\text{ ps}$. This indicates that the proposed method exhibits superior performance under conditions of temperature fluctuation.

VI. CONCLUSION

This article addresses the demands for high-precision time measurement by successfully developing an FPGA-based TDC

integrated with an online calibration method. This approach effectively tackles time resolution and linearity issues arising from the non-uniformity of internal delay elements in the FPGA. The research not only enhances the stability and accuracy of TDCs under varying environmental conditions but also provides a novel solution for time interval measurement in scientific research and engineering technology.

The core innovation of this work lies in utilizing calibration signals generated by an external high-precision clock to monitor changes in signal propagation time within the FPGA in real time. Two identical delay chains, managed by an FSM, alternate between primary and standby roles, ensuring continuous data acquisition and effective calibration.

Experimental results confirm that the proposed TDC significantly outperforms traditional TIC in terms of time linearity, INL, and DNL. Additionally, comparative experiments with the existing SR620 device validate the effectiveness of the proposed method, demonstrating comparable noise suppression capabilities. Both methods successfully maintain time difference fluctuations within $\pm 30\text{ ps}$. However, the proposed method exhibits minimal variation in measured time differences under temperature fluctuations of $\pm 1^\circ\text{C}$. This further affirms the efficacy of the online code density calibration method in mitigating the impact of temperature changes.

Overall, this research advances the theoretical development of high-precision time measurement technology and significantly enhances the reliability and robustness of the measurement system through innovative hardware design and intelligent algorithms. It lays a solid foundation for the application of TDCs in more complex environments.

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